L5.2: The null space of a matrix : finding nullity and a basis - Part 2

Transcript Language

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Mathematics for Data Science - 2 Professor Sarang Sane

Department of Mathematics, Indian Institute of Technology Madras

The null space of a matrix finding nullity and a basis - Part 2

So, let us do a three by four example. So, here your matrix is 1, 2, 0, 3, 2, 3, 0, 3 and 1, 1,

1, 2. So, in this example what do we want to do? We want to get what is the nullity

and a basis for the null space. That is what we want to do. So, the augmented matrix is this.

So, let us see what row reduction yields. So, row reduction is going to give me

1, 2, 0, 3, 2, 3, 0, 3, 1, 1, 1, 2. I wrote that matrix. And as I commented in my previous slide,

I am going to drop the augmented part. So, if we do that, we have swept out our first

column, everything below that one. Now, let us continue with the process. So, we get, we want a 1

in the 2, 2th place if possible. And indeed, we can multiply the second row by minus

1 and obtain that. So, we get 0, 1, 0, 3 in the second row, everything else remains the same.

So, let us continue the process. This is the same matrix we had in the previous

slide. So, now, let us sweep everything below the second,

in the second column below the 1. So, if you do that, we get 1, 2, 0, 3, 0, 1, 0, 3, 0, 0, 1, 2.

And this is already been reduced, not reduced in the row echelon form. But

we can go ahead one step and do reduced row echelon form. And if we do that, we get 1,

0, 0 minus 3, 0, 1, 0, 3, 0, 0, 1, 2. So, we have reached our matrix which is the

reduced row echelon form. Now, let us ask what are the dependent and independent variables.

The leading ones are in the first, second and third column that means x1, x2, x3 are dependent

variables. That means x4 is an independent variable. So, the independent variable is x4,

dependent variables are x1, x2, x3. So, the nullity of A is 1 right the nullity of A is 1,

because the nullity corresponds to the number of independent variables.

And what is the null space of A? So, for the null space, we put the independent variables to be the

tis. In this case, there is only one independent variable. So, instead of putting t1 I have put,

just called it t. So, put x4 equals t. So, the equations from the row reduced echelon

form are x1 minus 3x4 is 0. So, here I am using the augmented matrix. So, I have this matrix 1,

0, 0, minus 3, 0, 1, 0, 3 and 0, 0, 1, 2. And I am using the augmented matrix.

So, if I call this R, I am using the augmented matrix R0. And from there,

I am getting these equations. So, I get x1 minus 3x4 is 0, x2 plus 3x4 is 0,

and x3 plus 2x4 is 0. So, from here we get that x1 must be 3t by putting x4 as t, x1 must be 3t, x2

is minus 3t and x3 is actually minus 2t. So, now we know what is the general form of a solution.

It is of the form, so 3t, minus 3t, minus 2t and t and t varies over R. So, some possible

solutions are given by, let us say, if you put t equals 5, you get 15 comma minus 15 comma

minus 10 comma 5 or if you put a t equals 100, you get 300 comma minus 300 comma minus 200 comma 100

or you could of course put t equals 0 also. That is also a solution.

That is the trivial solution, 0, 0, 0, 0. So, actually, other than the last one the 0,

0, 0, 0, any of these we could have chosen as a basis vector. But just to follow a procedure,

we put t equals 1. So, t equals 1 yields a basis vector, 3, minus 3, minus 2, 1. So, a basis

for the null space of this matrix is just a single basis vector 3, minus 3, minus 2, 1.

So, you can check that this is actually inside your null space by multiplying this to A

on the right of course as a column and seeing that the result is 0.

Fine so, we have done these two examples of

computing the nullity of a matrix and the null space of a matrix and its

basis. So, now let us take a step back and and and ask what did we exactly do? So, we found a general

solution for the system of linear equations Ax equals 0. And then we put the ti s to be

1 and everything, all other tj s to be 0 and each such will give you

a basis vector. This is the main point, the main idea.

Fine so, let us do some observations and these observations are going to yield a theorem,

which is called the rank nullity theorem.

So, what is the rank nullity theorem? So, let A be an m by n matrix.

So, recall that the row rank of A is the dimension of the row space of A and the column rank of A is

the dimension of the column space of A. So, what do we mean by the row space of A?

It is the vector space or the subspace of Rn if it is an m by n matrix of Rn,

which is spanned by the rows of A, which we treat as vectors. And what is the column space of A? It

is a subspace, it is the subspace of Rm which is given as the span of the columns of A, meaning

you treat each column as a vector in Rm and that span is going to give you the column space.

So, recall that these are equal. This is something I stated, but did not prove and

they are denoted by rank of A. So, the rank of A is any one of these two, either the row rank

or the column rank, because they are both the same. So, we used in a previous video, we used

the fact that rank of A is the same as the row rank to compute this rank. How did we do that?

We said, you take this matrix A, you do row reduction, and once you do row reduction, you

look at the non-zero rows of this matrix. And the number of non-zero rows is exactly the rank of A.

So, recall this if you do not remember. So, now this is an important point. So, note that for

a matrix R in the row echelon form, so suppose you have a matrix which is in row echelon form,

the number of non-zero rows is the same as the number of dependent variables.

How do we get this? Each non-zero row has a leading one, because it is in row echelon form.

And each leading one that corresponding column corresponds to a dependent variable.

So, the non-zero rows correspond to the leading ones correspond to the columns with leading ones

correspond to the dependent variables. That is how this correspondence is. So,

the number of non-zero rows is the same as the number of dependent variables.

Of course, dependent variables were, because we there is no system of equations in this place. So,

as I said, here you think of the, in your mind, you think of the system of linear, the homogeneous

system of linear equations, Rx equals 0. So, you have this matrix A, you row reduce it to R,

think of the system of equations Rx equals 0, think of how many dependent variables there are,

that is exactly the same as the number of non-zero rows. That is what we are saying.

And what is the number of non-zero rows? That is the same as the rank of the matrix A.

Why is that the case by the way, because remember that in the previous video we have seen that,

if you have a spanning set for vector space, you put those in into your matrix as rows.

But in this case, we have already done that. That matrix is by construction that we started

with the rows are the vectors which span the row space. So, that is how we obtain the statement.

So, the moral of the story is, the rank of A is the number of non-zero rows of R, R was remember

obtained as, from A by doing row reduction, which is the number of dependent variables of Rx is 0.

How is this related to nullity? Remember that the nullity of A is the number of independent

variables of Rx is 0. So, rank is the number of dependent variables,

nullity is the number of independent variables. But well, if you have a variable either it is a

dependent variable or an independent variables, so the number of dependent variables plus the

number of independent variables is the total number of variables which is exactly n,

remember, it is an m by n matrix. So, this gives us the rank nullity theorem

that the rank of A plus the nullity of A is n. Why, because the rank is a number of

dependent variables and the nullity is a number of independent variables and we just saw that

the sum is the total number of variables which is n. So, this is a

heuristic proof of the, or I let me remove the word proof, this is a heuristic for this theorem.

So, this is a useful and and often quoted theorem in linear algebra.

So, with that, let us do the final topic that we want to do in this video. How do we check if a set

of n vectors is a basis for Rn? So, remember that for Rn we have a standard basis. Namely,

the eis which is 1 in the i-th place, i-th coordinate and 0 is everywhere else.

This is the standard basis for Rn. And we know that the size of any basis is the same.

So, the, that was what we had called dimension. So, we know that the dimension of Rn is n.

So, we know that if at all you have a basis it must consist of n vectors. So,

now suppose we are given n vectors, how do I check whether or not this is a basis?

So, here, we use the idea from a previous

video about linear independence and dependence. Namely, that if you have a set of vectors and you

want to check whether or not they are linearly independent, so if you have, for example,

n vectors in Rn, you look at the determinant and that is exactly what we are going to do now.

So, that will that will tell you whether or not they are linearly independent.

But then the point is if you have, if they are independent, when you have n linearly

independent vectors and they must form a basis, because if not, once there will be some vector

which is not in the span of these vectors which is linearly independent so you add that, and you

can keep doing this process as we did. And in the end you, after appending a bunch of vectors,

you will get a basis. But then remember that the dimension will be strictly bigger than n.

So, the only way that this is possible is if your original set was itself a basis. So,

what is the moral? The upshot is that if you have n vectors, it is enough to check that

they are linearly independent. Other way of doing this is to say it is enough to check

that there are a spanning set. In either of these two cases, there must be a basis.

So, how do I check that in this case? In this case, it so happens that since you have n and n,

the corresponding matrix is a square matrix and I can look for its determinant.

So, let us let us see how to check this. So, the short answer is use determinants. And I

sketched out how. Suppose we are given n vectors of Rn, we write them as columns of a matrix.

So, we will obtain an n by n square matrix.

If the determinant of the matrix is 0, then the given set of vectors does not form a basis,

otherwise it forms a basis. So, what it saying is that in a very large number of cases,

almost always, which has a precise meaning, but we would not say what that is now, almost always,

they do form a basis. So, what do you have to do to check? You have to compute its determinant.

So, let us do examples. The quick examples are in R2, you take the standard basis, 1,

0, and 0, 1. We already know it is a basis. So, let us just check that the statement is correct.

So, what is the corresponding matrix? It is the identity matrix.

The determinant is 1. And indeed, this forms a basis. So, the determinant is non-zero.

And let us look at the vectors let us say 1, minus 2 and 5, minus 10.

So, this yields a matrix 1, minus 2 in the first column, 5, minus 10 in the second column.

And if you look at its determinant, it is 0. Of course, we know this, because they are

linear multiples of each other. And as a result, this is not linearly independent

and so it cannot form a basis. Let us do this for a slightly more

larger example and maybe a more involved example. So, is the set 1, 2, 3, 0, 1, 2, 1, 3, 0 a basis

for R3? Let us form the matrix. So, form a matrix A with columns given by these vectors, so that is,

the first column is 1, 2, 3, the second column is 0, 1, 2, the third column is 1, 3, 0.

Let us find its determinant. Well, we know how to do this.

So, I am, I have just directly done this. So, it is 1 times minus 6, minus 0 times minus 9,

which will anyway does not count, plus 1 times 4 minus 3. So, that is minus 6 plus 0

plus 1, which is minus 5, and that is non-zero. So, this does give you a set of vectors which form

a basis. Let me directly show you that this is indeed a basis by proving that this is a spanning

set for R3. So, let B belong to R3. So, I want to get a bunch of scalars

such that a1 times, if I call these vectors x1, x2, x3, so this is x1, this is x2, this is x3.

So, I want to get, so need a1, a2, a3 belonging to R such that summation aixi is B. I want this.

So, in other words, I want to solve A times a1, a2, a3 is equal to B. Why is that,

because the first column remember is 1, 2, 3, which corresponds to x1, the second column is

0, 1, 2, which corresponds to x2 and the third column is 1, 3, 0, which corresponds to x3.

So, if you write out this equation in terms of the columns, you will exactly have a1x1 plus a2x2 plus

a3x3. So, we want to solve this for a1, a2, a3. But now we know A is invertible.

So, we know a solution. In fact, there is a unique solution. So, unique solution is

a1, a2, a3 is equal to A inverse b. And why do we, I mean, where am I getting A inverse from,

well, because the determinant is non-zero. So, the inverse exists.

In fact, we know explicitly how to compute it. The point is that there is a solution. So,

that means a1, a2, a3 exists. So, scalars a1, a2, a3 exists

such that summation aixi is equal to B for any B in R3. That means it is a spanning set for R3.

Already, the fact that determinant is non-zero tells you it is linearly independent

that means it is a basis. So, that is why it is enough to check for the determinant.

Let us do one last example in R4. So, let us have these four vectors 1, 2, 3, 0, 0, 1, 2, 1, 1, 3,

0, 2, 2, 6, 5, 3 is this a basis for R4? What do I do? I make the matrix with these columns. Put

these vectors in the columns. So, in the first column, I have 1, 2, 3, 0. In the second column,

I have 0, 1, 2, 1. In the third column, I have 1, 3, 0, 2. In the fourth column, I have 2, 6, 5,

3. Compute its determinant. We know how to compute determinants.

The, we will do this by, well, the easiest way would have been by doing Gaussian elimination

or row reduction rather, but let me do it by hand. So, by hand,

let us do it by expansion along the first row. So, I get 1 times the determinant of the 1, 1 minor,

minus 0 times the 1, 2 minor, plus 1 times the 1, 3 minor, minus 2 times the 1, 4 minor. So,

the 0 times part I have removed, because anyway I am going to get the, that will contribute nothing

and the rest are written down here. So, you can compute this. The first term is 11,

second is 0, so I have removed it altogether, the third term is 11 and the fourth term is 2 times 11

and so this is 0. And as a result, the given set of vectors do not form a basis of R4.

Thank you.